

Literature Status: A Complete Combinatorial Classification of Bounded Slice Spaces

fa_banach_001, lane 8

August 9, 2026

Status. `literature_already_answered`. This is an exact answer explicitly identified by the supporting authors, not an agent-inferred implication.

Original source and question. Gaubert and Gunawardena, *The Perron–Frobenius Theorem for Homogeneous, Monotone Functions*, arXiv:math/0105091, later Trans. Amer. Math. Soc. 356 (2004), 4931–4950. In the subsection “Slice spaces and recession functions,” journal page 4937, after giving a sufficient recession-function condition, they ask:

It remains an open problem whether the boundedness of all slice spaces can be determined by combinatorial or graph-theoretic constructions as in Theorems 4 and 6.

Here a slice space is

$$\mathcal{S}_\alpha^\beta(f) = \{x \in \mathbb{R}_{\geq 0}^n : \alpha x \leq f(x) \leq \beta x\}, \quad \alpha, \beta > 0,$$

for a monotone positively homogeneous self-map f of $\mathbb{R}_{\geq 0}^n$, and boundedness is measured in Hilbert’s projective metric.

Supporting answer. Akian, Gaubert, and Hochart, *A game theory approach to the existence and uniqueness of nonlinear Perron–Frobenius eigenvectors*, arXiv:1812.09871. Its introduction restates the source question verbatim in substance as Problem 1 [GG04, p. 4937] and says “we ... solve Problem 1.” The exact answer is Theorem 1.2: all slice spaces of f are bounded in Hilbert’s projective metric if and only if the two players have no disjoint dominions in the finite-state game $\Gamma_\infty(f)$. The legal actions of this game are defined only from coordinatewise growth of the logarithmic conjugate $T = \log \circ f \circ \exp$ at $+\infty$ and $-\infty$, so this is a combinatorial criterion. Section 4 further shows that the dominion condition is decidable by reachability in two directed hypergraphs.

Identification and scope. The supporting paper cites the original article, gives the exact original page and problem, and explicitly presents Theorem 1.2 as the full characterization solving it. Thus the finite-dimensional slice-space problem is completely answered. This identification does not settle the distinct question, also raised in the 2004 article, of extending these methods to general cones in Banach spaces.

Evidence searched. The run indexes contained no prior record for this paper or question. An arXiv title-and-keyword search located arXiv:1812.09871; its abstract, Problem 1, Theorem 1.2, and Section 4 provide the decisive evidence.

References

- [1] S. Gaubert and J. Gunawardena, *The Perron–Frobenius Theorem for Homogeneous, Monotone Functions*, arXiv:math/0105091; Trans. Amer. Math. Soc. 356 (2004), 4931–4950.
- [2] M. Akian, S. Gaubert, and A. Hochart, *A game theory approach to the existence and uniqueness of nonlinear Perron–Frobenius eigenvectors*, arXiv:1812.09871.